

Load Interconnection System Impact Study Report

For Queue Project Q14-0402

Orangedale 138 kV

February 15, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0402. The Interconnection Customer proposes a 216.7 MW large load (data center) facility interconnecting at the Orangedale 138 kV substation in the Maryland area. The request was received on November 08, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 228 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,186,000.

3. Network Impacts

3.1 Load Deliverability Analysis

Gum Springs Substation to Carnestown 138 KV Line: 111.0% loading (206 MVA) following the outage of the Dc Water Chp to Jefferson Street 138 KV; DFAX = 7.1%.

The Mottman to Bloomington 138 KV #2 is overloaded to 109.5% of its 378 MVA emergency rating for the loss of the Mottman to Bloomington 138 KV #1. The project contributes 6.3% DFAX (13.6 MW) to this overload.

The Chancellor Row to Bear Island Substation 345 KV is overloaded to 105.6% of its 1,105 MVA emergency rating for the loss of the Birchwood to Leavitt 345 KV. The project contributes 5.5% DFAX (12.0 MW) to this overload.

For the loss of Dc Water Chp to Jefferson Street 138 KV, the Gum Springs to Grantline-Siefker 138 KV Line loads to 104.5% of its 233 MVA rating. Project DFAX: 10.1%.

For the loss of Birchwood to Leavitt 345 KV, the Bear Island to Dunn Siding 345 KV line loads to 100.7% of its 1,570 MVA rating. Project DFAX: 6.1%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Gum Springs Substation to Carnestown 138 KV Line	Rebuild line with higher-capacity conductor	\$1,468,000	10% (\$142,000)
Mottman to Bloomington 138 KV #2	Reconductor line	\$552,000	27% (\$151,000)
Chancellor Row to Bear Island Substation 345 KV	Reconductor line	\$13,238,000	2% (\$243,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Gum Springs to Grantline-Siefker 138 KV Line	Reconductor line	\$2,171,000	17% (\$360,000)
Bear Island to Dunn Siding 345 KV L	Reconductor line	\$21,931,000	1% (\$290,000)
	Total allocated to project		\$1,186,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.